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Tue, May 12, 2026 at 10:29 AM

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Track Name: Artificial Intelligence and Data Science

Paper ID: 1917

Paper Title: Custom Packaging Optimization for Cardboard Waste Reduction Using AI-Driven Techniques

Abstract:

The exponential growth of online retail operations has led to massive amounts of packaging waste caused by inadequate box size design and overprotection of goods during shipping. Consequently, high logistical and environmental impacts have been incurred. To mitigate such issues, an AI-powered optimization scheme for generating custom-fit packaging designs is presented in this study. In particular, the proposed scheme uses a clustering algorithm that is based on machine learning to divide products into clusters according to their similarities regarding physical properties. Then, a dynamic programming algorithm is used to find the optimal box size parameters and minimize unnecessary material consumption. Constrained optimization is introduced to account for real-world limitations, such as product protection and weight distribution.

To evaluate the proposed algorithm, a dataset with 8,947 unique items was created and analyzed using the proposed and baseline algorithms. Statistically, the algorithm outperformed the baseline fixed-margin approach by significantly reducing material usage and carbon footprint, which is associated with lower costs. Overall, the results demonstrate the effectiveness of artificial intelligence, optimization approaches, and informed decision-making in facilitating sustainable packaging solutions for contemporary e-commerce operations.

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Secondary Subject Areas: Not Entered

Submission Files:[Plagiarism_report.pdf](#) (265 Kb, Tue, 12 May 2026 04:54:26 GMT)[Packaging_Optimization_ResearchPaper.pdf](#) (4 Mb, Tue, 12 May 2026 04:55:39 GMT)**Submission Questions Response:**

1. Corresponding author Email ID
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+91 9041629434
3. Author's Information
Agreement accepted
4. Co-author's Consent
Agreement accepted

5. Certificate of Originality

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6. Author's Consent

Agreement accepted

7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

Agreement accepted

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